

Here is a unique puzzle game idea that fits your criteria for **Minimalist design, mathematical elegance, and an addictive core loop**, called **"Factor Field."**

Game Idea: Factor Field (The Prime Grid Puzzle)

Success Factor	Implementation
Minimalist Design	A simple square grid of black and white tiles; clean, unadorned numbers.
Mathematically Elegant	Based purely on Prime Factorization and divisibility.
Addictive Core Loop	Simple selection mechanic with complex, cascading consequences.
High Achievement	Solving the board requires complex spatial and numerical deduction.

The Core Loop

1. The Board

- The game is played on an $N \times N$ grid (e.g., 5×5 or 7×7).
- Each tile contains a single, non-prime integer (e.g., 4, 6, 8, 9, 10, 12, etc.).

2. The Goal

- The player must select tiles until **every tile on the board has been reduced to the number 1.**

3. The Mechanic: Prime Selections

- The player **selects a single Prime Number** from a small, limited palette provided for the current level (e.g., Primes 2, 3, 5).
- **The Action:** Upon selecting a Prime P , the following happens simultaneously:
 - **Division:** Every tile on the entire grid whose number is **divisible by P** has its value instantly divided by P .

- **Chain Reaction:** The new values trigger new potential divisibility for the next selection.

4. The Puzzle

The challenge lies in the **order of operations** and the **limited resource** (the Prime palette):

- **Optimal Order:** A single mistake in choosing the Prime sequence can leave a single tile stranded with a value that is no longer divisible by any remaining Primes in the necessary combination.
 - *Example:* If a tile has the value 9, you must use Prime 3 twice. If you use Prime 2 too many times early on, you might run out of valid moves to clear other tiles, leaving the '9' stranded.
- **Minimalist Win:** The player aims to solve the entire board using the **minimum possible number of Prime selections**.



Mathematical Elegance & Achievement

The game is a pure exercise in applying the **Fundamental Theorem of Arithmetic** (every integer greater than 1 is either a prime number itself or can be represented as a product of prime numbers).

- **Addictive Loop:** The instant, grid-wide cascading division provides satisfying feedback, making the player eager to see the next move's consequence.
- **High Achievement:** Solving a level is proof of having found the single, correct **sequence of prime divisions** that clears the board to 1 while conserving moves. For larger boards, the number of possible sequences grows exponentially, making a solution feel mathematically elegant and earned.

Example Move:

Board State (Start)	Prime Selection	Board State (End)
8, 10, 9	Select Prime 2	4, 5, 9
6, 4, 12	Select Prime 3	4, 5, 3
15, 6, 25	Select Prime 5	3, 5, 5